High-Volume Error Attack

M5 Retail Demand Forecasting — Problem Statement 11

Phase 3 — diagnosis and three failed fixes

NPN AIA Hackathon — St. Joseph's College of Engineering

Phase 3 — where the error really is, and three attempts to fix it. Generated 2026-08-14.

> ***Terms. RMSE** — average error with big misses punished far more heavily; lower is better. **MAE** — the plain average error. **Leakage** — letting the model see information that would not have existed when the forecast was really made. **Tweedie** — a loss function for data that is never negative and mostly zero. **Objective / loss function** — the definition of "wrong" that the model tries to minimise. **Inner window** — an earlier 28-day period we tune on, so the real scoring window stays untouched.*

> *Every number here comes from an experiment that actually ran. Failed experiments are reported alongside successful ones — in this campaign most of them failed, and that is the finding.*

Diagnosis: where RMSE comes from

Volume tier	Rows	Actual mean	Predicted mean	Bias	RMSE	Share of squared error
high	65,968	7.585	7.196	-0.389	5.976	61.33%
medium	160,244	2.185	2.076	-0.109	2.291	21.90%
low	398,160	0.797	0.761	-0.036	1.174	14.30%
very low	229,348	0.280	0.267	-0.012	0.643	2.47%

The busiest tier is 7.7% of rows and carries about 61% of all squared error. We under-predict it by roughly 0.39 units per row.

Concentration is even sharper by product: the **top 50 items of 3,049 carry 37.7% of all squared error**.

Three legitimate fixes, all tested

Attempt	RMSE	MAE	∆RMSE	High-volume RMSE	High-volume bias
current best (reference)	2.1210	1.0319	+0.0000	5.9756	-0.3890
volume weight cap 3.0x	2.1376	1.0336	+0.0165	6.0530	-0.4770
volume weight cap 5.0x	2.1371	1.0335	+0.0161	6.0484	-0.5054
high-vol calibration x1.00	2.1210	1.0319	+0.0000	5.9756	-0.3891

All three failed, and the way they failed is informative

Volume weighting made things worse. Weighting busy series more heavily in training pushed RMSE up by about 0.016 — and, counter-intuitively, made the high-volume tier itself *worse* (RMSE 6.05 versus 5.98) with a *more* negative bias. Forcing the model to chase big days cost it accuracy on the medium ones without buying accuracy on the big ones.

Post-hoc calibration found nothing to correct. We searched for a multiplier to apply to high-volume predictions, choosing it on the inner window. The search returned **1.00** — no scaling improved anything. The model is already optimally calibrated for that tier.

What we conclude

The under-prediction of busy days is **not a calibration error and not a weighting error**. It is what a squared-error-family model correctly does when the target is genuinely volatile: predicting the conditional mean of a high-variance day is the right answer even though it looks timid. The remaining high-volume error appears to be irreducible with the information available.